

Recognise and use rules that generate visually growing patterns and number patterns involving rational numbers

Learning intention

- recognise and use rules that generate visually growing patterns and number patterns involving rational numbers.

Teach and model

- investigating patterns such as the number of tiles in a geometric pattern.
- creating an extended number sequence that represents an additive pattern using decimals.
- using a calculator or spreadsheet to experiment with number patterns that result from multiplying or dividing.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.